

Mean–Variance Dynamics and Market Efficiency in Silver

A multi-model forecasting framework — weekly silver returns and volatility, 2015–2026

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Q1 · WEAK FORM

NO

Are weekly returns forecastable from their own past?

AIC picks ARIMA(0,0,0) — the drift itself; the spectral test can’t reject the martingale.

Q2 · SEMI-STRONG

NO

Does public information help — linear or nonlinear?

0 of 16 model selections beat the drift out of sample (smallest DM p = 0.43).

Q3 · THE VARIANCE

YES

Is weekly realised volatility forecastable?

Every model beats the naïve floor; HAR alone cuts its mean-squared error by a third.

Q4 · SENTIMENT

RISK ONLY

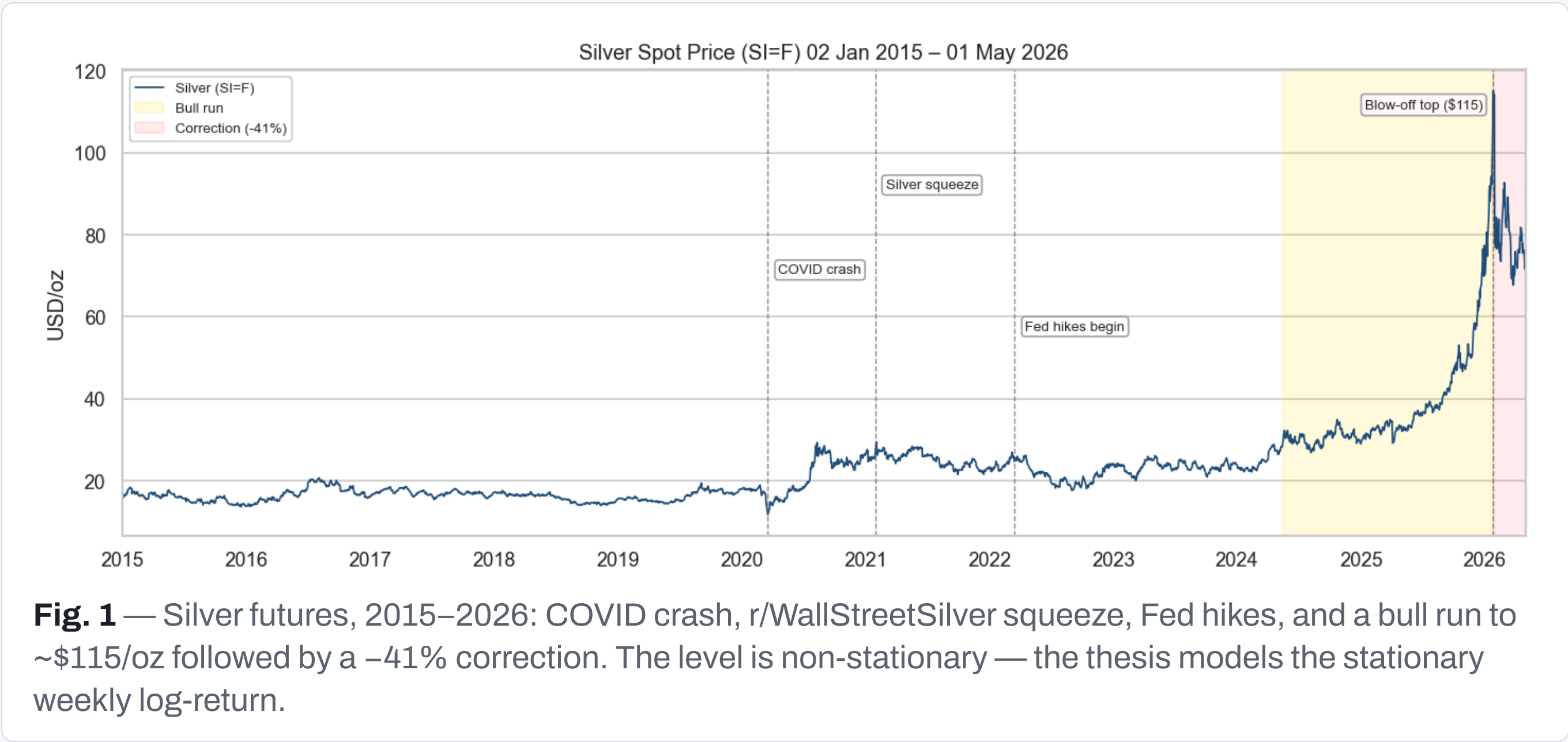
Does Reddit & news sentiment add anything?

Nothing to the mean; a small but robust gain to volatility (QLIKE p < 0.01).

The mean is a *martingale*; the variance is *not*.
Predictable risk alongside an unpredictable return is exactly what an efficient market permits — one statement, not two studies.

01 One series, two moments

591 Friday-to-Friday weeks · target $r_t = \log P_t - \log P_{t-1}$



Silver lives a double life — an industrial input (~half of demand: photovoltaics, electronics), a monetary hedge, and an episodic meme trade (spot \$25→\$30 in days, Jan 2021). An unusually rich laboratory for the oldest question in empirical finance: *how much of a price can be forecast?*

Returns: almost no autocorrelation — a linear model has nothing to grip.
Squared returns: strongly autocorrelated — volatile weeks cluster.
The structure is not gone; it lives in the second moment.

The weekly horizon is where the information aligns: COT is weekly, macro monthly, daily returns are microstructure noise — and ~600 observations remain to train and honestly test the ML models.

02 Six source families, one weekly frame

every feature lagged before use — no look-ahead

Market prices
Yahoo · daily — silver + 6 cross-assets

Rates & inflation
FRED · daily — real yield, breakevens, claims

Macro releases
FRED · monthly — point-in-time rule

Positioning
CFTC COT · weekly — funds vs hedgers

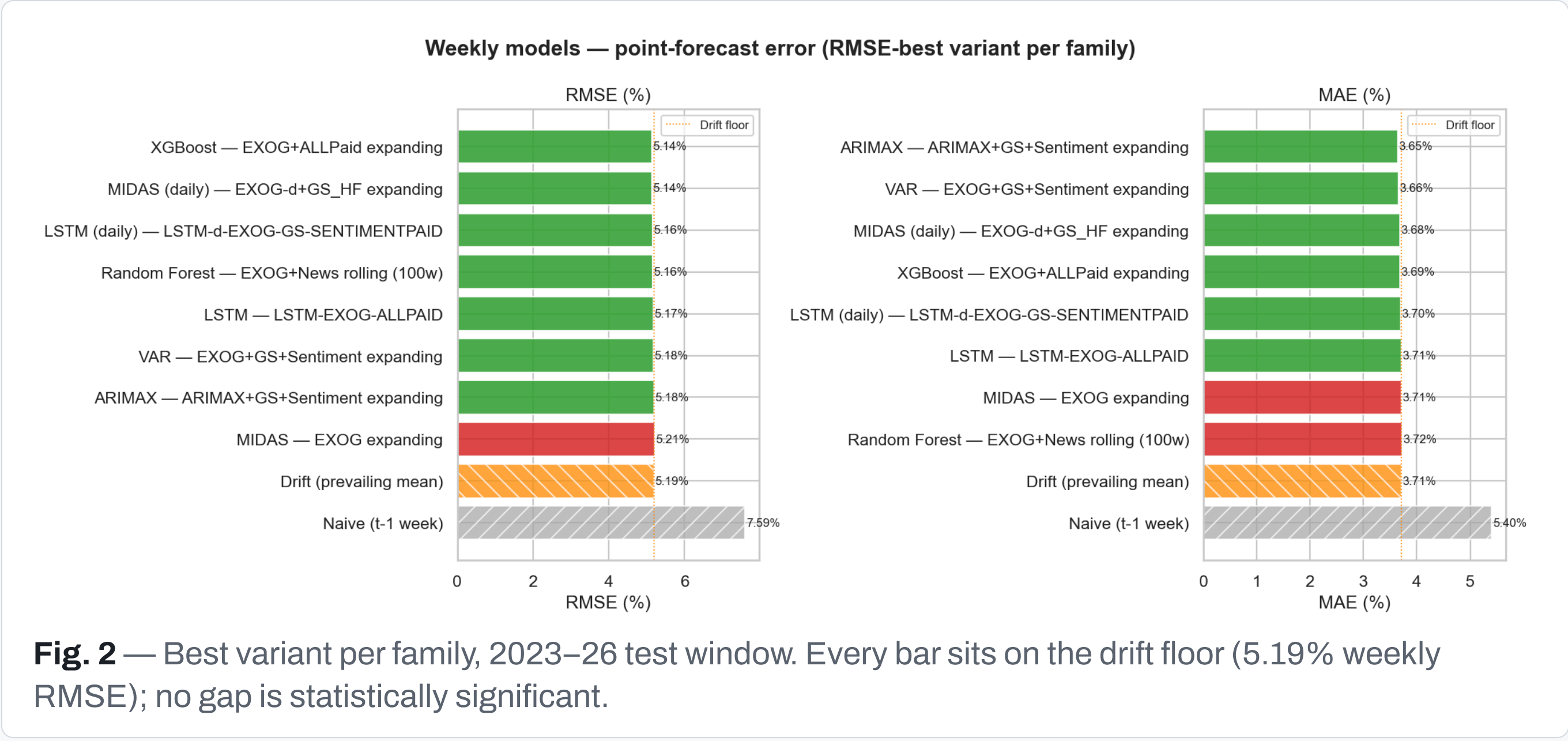
Sentiment
Reddit · news · Trends — RoBERTa + FinBERT

China PMI
FRED · monthly — industrial demand

591 weeks → **Train 2015–21** 365 wk **Val '22** **Test 2023–26** 174 wk

03 The mean: nothing beats the drift

ARIMA(X) · VAR · MIDAS · RF · XGBoost · LSTM — vs random-walk-with-drift, walk-forward



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family selections beat the drift on a Diebold–Mariano test

0.43

smallest DM p-value across every family and loss

+1.8%

best out-of-sample R^2 (XGBoost) — within noise

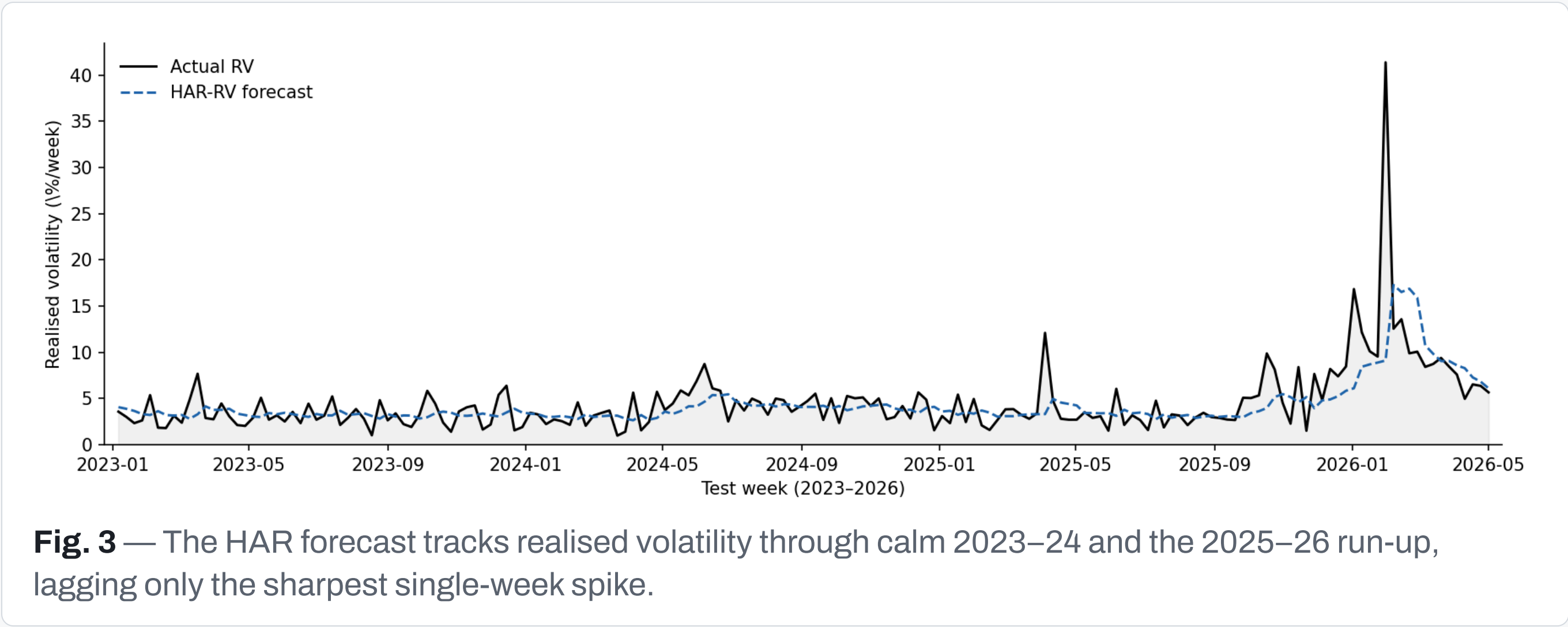
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sub-periods where no family holds a stable edge — regime luck, not skill

A directional footnote. Linear models time the sign (Pesaran–Timmermann $p \approx 0.005$) — but the signal is magnitude-worthless (negative R^2_{OS}), concentrated in the 2025 bull, and absent from the nonlinear families. An adaptive-markets curiosity, not a rejection of efficiency.

04 The variance: strongly forecastable

weekly realised volatility · GARCH · HAR · tree ensembles — vs naïve RV_{t-1}



–1/3 MSE

vs the naïve floor — $R^2_{OS} = 0.29\text{--}0.32$, all models

71%

of weekly volatility rises & falls called correctly (HAR)

p < 0.01

QLIKE Diebold–Mariano vs naïve — every model

Forecast error, RV points (RMSE) — lower is better

XGBoost (HAR)	3.13
Random Forest (HAR)	3.15
HAR-RV	3.20
GARCH(1,1)	3.28
Naïve (RV_{t-1})	3.99

Cross-asset volatility spillover **hurts** (HAR+EXOG loses to bare HAR, $p \approx 0.03$): silver’s forecastable variance is its own, not gold’s or copper’s. Collapsing HAR to one lag significantly degrades the forecast — the signal is multi-horizon persistence.

05 Attention prices risk, not direction

06 Does it matter? Ask a risk manager

Does a text signal improve HAR? QLIKE DM statistic — left of zero improves

Reddit sentiment	–3.36 ***
Paid-news attention	–3.33 ***
Reddit both	–3.01 **
Reddit attention	–2.60 **
Google Trends	–0.01 (tie)
Paid-news sentiment	+1.44
Cross-asset spillover	+2.23 * worse

Linear HAR runs, 174 common weeks. **How much is written** beats **how it reads**: attention and tone-intensity help; polarity alone does not. The same block never helps the mean.

One-week 5% Value-at-Risk backtest breach should sit near 5% · pinball $\times 10^4$, lower is better

VOLATILITY FORECAST	BREACH	PINBALL
Naïve (RV_{t-1})	8.0%	66.3
GARCH(1,1)	3.4%	58.5
HAR	6.3%	59.6
HAR + Reddit attention	5.7%	59.6

The HAR-based VaR is well calibrated and beats the naïve floor at **p < 0.001**; the naïve VaR over-breaches and overshoots after every spike. Forecastable volatility is the difference between a usable and an unusable tail-risk number — position sizing, option pricing, and risk budgets all consume the *variance*, not the mean.

One ladder, one benchmark
Classical, ML and deep models judged on a level footing, for both moments.

Explicit factorisation
“Returns failed, volatility worked” becomes one coherent efficiency statement.

Domain-tuned sentiment
RoBERTa for Reddit, FinBERT for news — tested as a falsifiable ablation.

Qualified negative results
Direction separated from magnitude; apparent signals traced to regime luck.